

Robotics in Motion

From sensor inputs to coordinated legged movement

Qihang Lu | Robotics summer programme: Huzhou and UCL

During this summer programme, I explored how mechanical structures, sensors, and control code work together to produce robot movement. Preparatory training at the Huzhou Institute of Zhejiang University, from 30 July to 1 August 2025, was followed by laboratory visits and practical robotics sessions at UCL in early August.

Building a foundation in control

In Huzhou, I worked with digital and analogue inputs, servos, motors, and ultrasonic sensors. I assembled a transport robot and organised its movement commands into reusable functions, adjusting motor speeds, movement durations, and sensor thresholds during testing. I also learned the principles of PID feedback control. For a joystick-control exercise, I handled the client-side reception of data while a classmate developed the server side, allowing us to establish basic remote control together.

Developing and refining a walking robot

At UCL, I assembled a small legged robot and examined how coordinated hip and ankle movements create a walking cycle. Translating this sequence into reliable motion required repeated adjustment: the feet sometimes collided, friction and weight distribution prevented a foot from lifting, and motor problems required disassembly and recalibration. I tuned a four-step walking sequence and then upgraded it to a more stable six-step gait. This process showed me how closely code parameters depend on the physical assembly.

Connecting perception with movement

After testing sensor outputs, I developed flowcharts and substantially completed the code for target-following and obstacle-avoidance modes. The logic included stopping when a target was too close and reversing near an obstacle to create room for turning. In the final navigation exercises, I adjusted gait parameters for the test environment and used sensor readings to guide movement. The robot finished near the intended destination, although its positioning remained imprecise.

What I took away

Visits to UCL robotics laboratories introduced me to humanoid and hybrid wheeled-legged systems, alongside challenges in sensing, manipulation, and reliable testing. Building and troubleshooting my own robot made these challenges tangible. The experience strengthened my interest in mechanical engineering, especially the coordination of mechanisms, actuation, and sensing that allows robots to move reliably in their surroundings.